

VxLEARN Networks

Networking & Cybersecurity Track
Simulated Employment Program

Lab Report: File Data Integrity Checks

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Objectives

- Part 1: Recover Files after a Cyber Attack
- Part 2: Using Hashing to Verify File Integrity
- Part 3: Using HMAC to Verify File Integrity

Background / Scenario

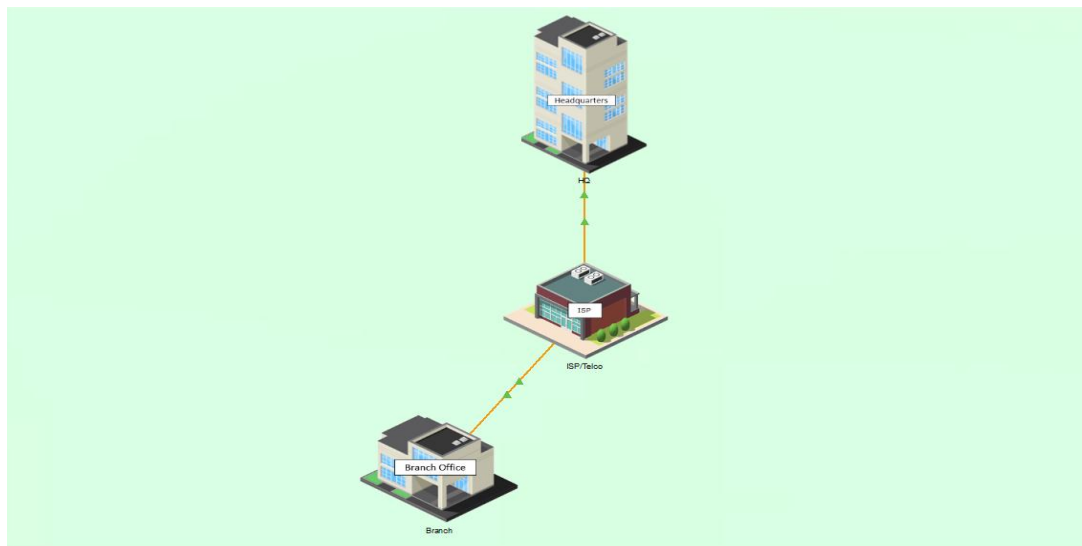
In this Packet Tracer (PT) activity, you will verify the integrity of multiple files using hashes to ensure they have not been tampered with. If any files are suspected of being tampered with, they are to be sent to Sally's PC so that she can further analyze the contents. The IP addressing, network configuration, and service configurations are already complete. You will use the client devices to verify and transfer any suspect files.

Key concepts applied in this lab: MD5 hashing, SHA-256 HMAC, FTP file transfer, incident response workflow, and the use of Linux command-line tools on the CSE-LABVM.

Network Topology

The lab consists of two sites (Branch Office and HQ) with the following key devices:

- Laptop BR-1 - Mike's PC (Branch Office): accesses branch.corp and hq.corp
- HQ-Laptop-1 - Sally's PC (Headquarters): security analyst / incident responder
- HQ-Laptop-2 - Bob's PC (Headquarters, CFO): stores the critical file income.txt
- BR Server - Branch web/file server (<http://branch.corp>)
- HQ FTP Server - Backup and archive server (<ftp://hq.corp>)
- CSE-LABVM - Linux virtual machine (VirtualBox) used for hashing operations

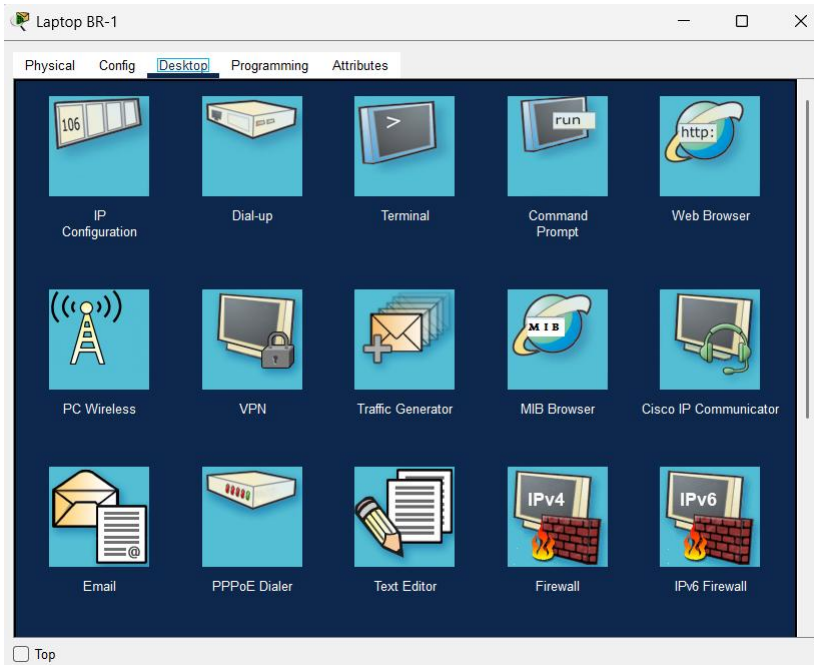


Part 1: Recover Files after a Cyber Attack

Client data must be secured and remain unchanged by unauthorized personnel. By hashing data before and after it is archived, you can tell if it has changed even by one character or one bit, because the hashes will not match. In this part, you will attempt to recover files from a backup after a cyber attack.

Step 1: Access the BR Server from Mike's PC

a. In Packet Tracer, click Branch Office and then click Laptop BR-1.

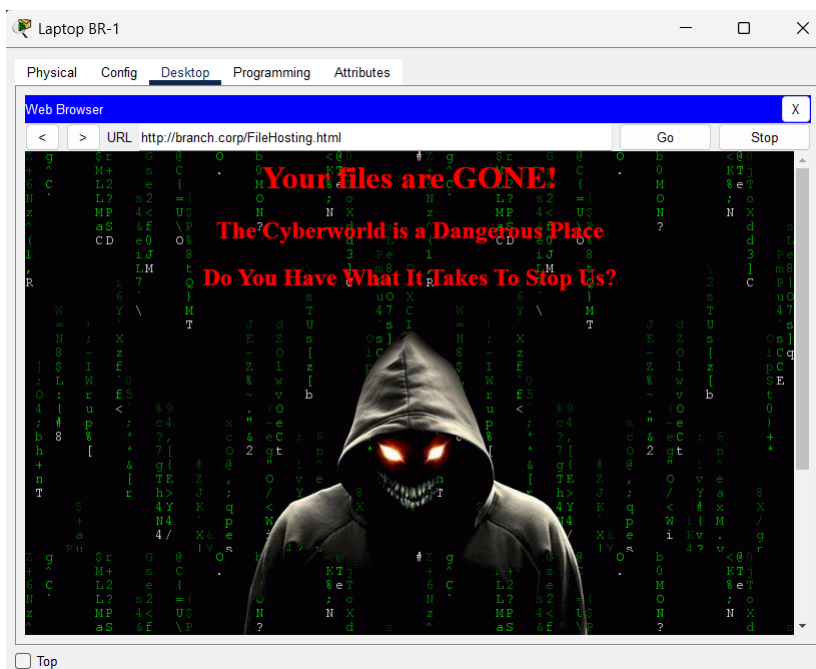
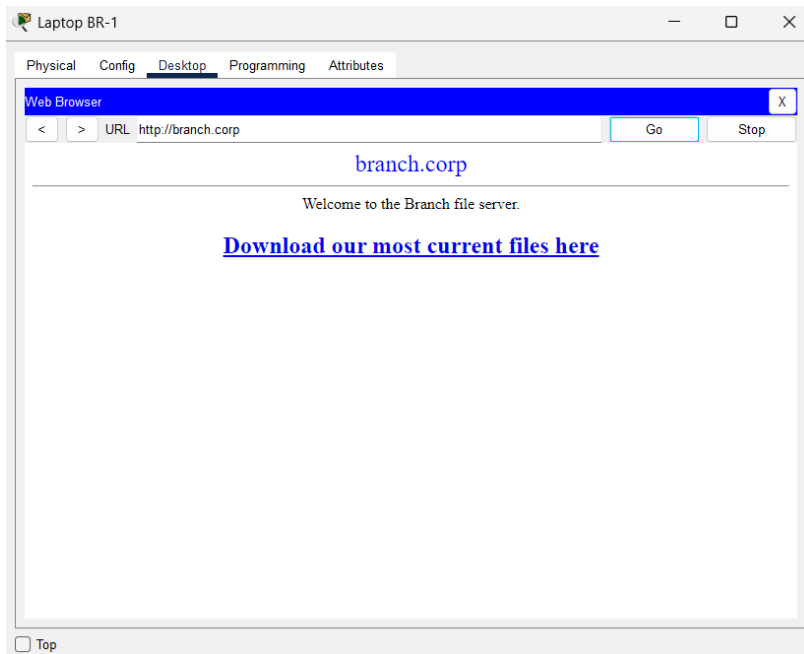


b. Click the Desktop tab, then click Web Browser.

c. In the address bar, enter `http://branch.corp` and click Go.

Note: Packet Tracer may take up to a minute to converge. Press Alt+D (Fast Forward Time) to speed up convergence.

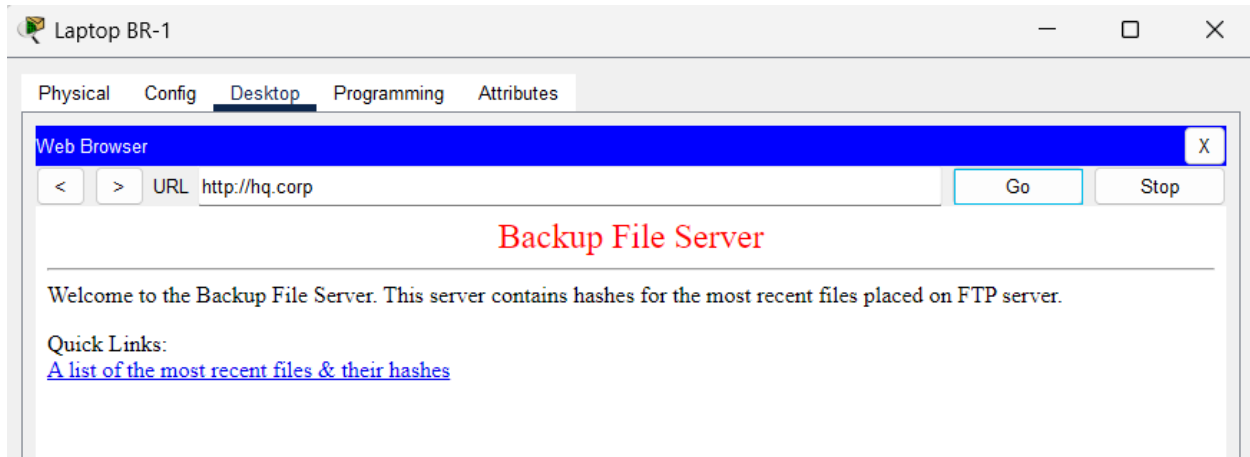
d. Once the page loads, click the link to download the most current files.



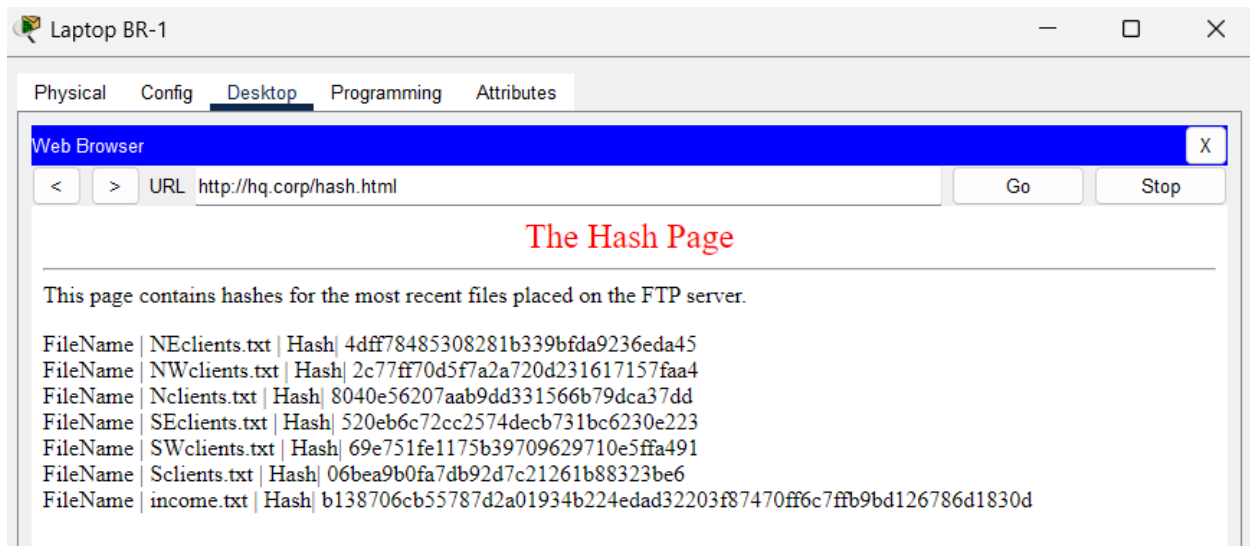
Step 2: Copy the Hash Values from the Last Archive

You need to restore the missing files from the HQ backup server. First, obtain the archived hash values that will be used to validate the integrity of recovered files.

a. In the Web Browser, navigate to <http://hq.corp> and click Go.



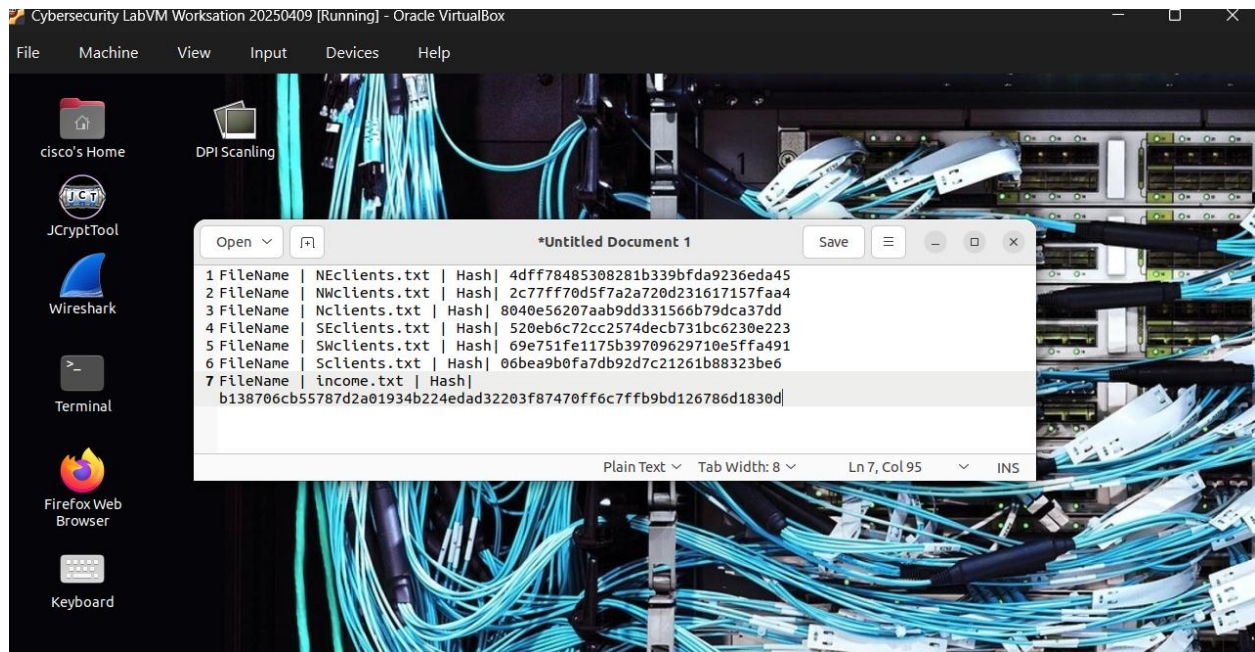
b. Click the link to view the most recent files and their corresponding hash values.



c. Select all the content on the page and copy it (Ctrl+A, Ctrl+C).

d. Open the CSE-LABVM, then click Menu > Text Editor Pluma.

e. Paste the clipboard contents into the blank Pluma document (Ctrl+V). This document will serve as your reference list of archived hash values.

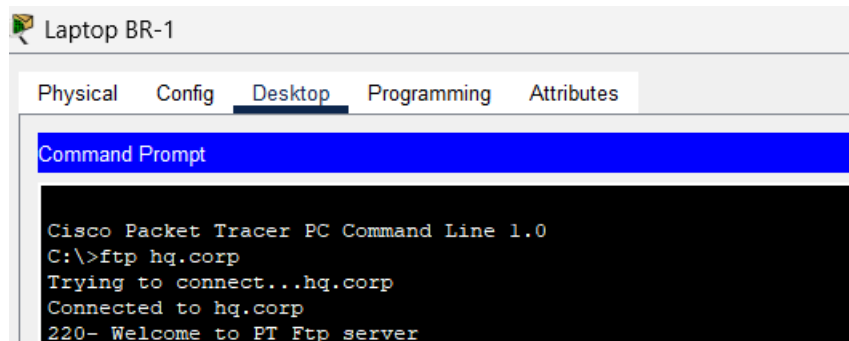


Step 3: Download the Backup Files to Mike's PC via FTP

a. Back in Packet Tracer, close the Web Browser on Mike's Laptop BR-1.

b. Click Command Prompt. Connect to the HQ FTP server:

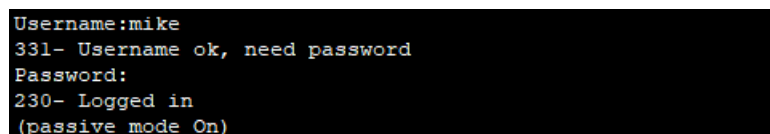
```
ftp hq.corp
```



c. Log in when prompted:

```
Username: mike
```

```
Password: cisco123
```



d. At the ftp> prompt, enter dir to list the available files on the server:

```
ftp> dir

ftp> dir

Listing /ftp directory from hq.corp:
0   : NEclients.txt           582
1   : NWclients.txt           583
2   : Nclients.txt            698
3   : SEclients.txt           597
4   : SWclients.txt           649
5   : Sclients.txt            780
```

e. Download all six client files using the get command:

```
ftp> get NEclients.txt
ftp> get NWclients.txt
ftp> get Nclients.txt
ftp> get SEclients.txt
ftp> get SWclients.txt
ftp> get Sclients.txt

ftp> get NEclients.txt

Reading file NEclients.txt from hq.corp:
File transfer in progress...

[Transfer complete - 582 bytes]

582 bytes copied in 0.044 secs (13227 bytes/sec)
ftp> get NWclients.txt

Reading file NWclients.txt from hq.corp:
File transfer in progress...

[Transfer complete - 583 bytes]

583 bytes copied in 0.03 secs (19433 bytes/sec)
ftp> get Nclients.txt

Reading file Nclients.txt from hq.corp:
File transfer in progress...

[Transfer complete - 698 bytes]

698 bytes copied in 0.014 secs (49857 bytes/sec)
ftp> get SEclients.txt

Reading file SEclients.txt from hq.corp:
File transfer in progress...

[Transfer complete - 597 bytes]

597 bytes copied in 0.016 secs (37312 bytes/sec)
ftp> get SWclients.txt

Reading file SWclients.txt from hq.corp:
File transfer in progress...

[Transfer complete - 649 bytes]

649 bytes copied in 0.064 secs (10140 bytes/sec)
ftp> get Sclients.txt

Reading file Sclients.txt from hq.corp:
File transfer in progress...

[Transfer complete - 780 bytes]

780 bytes copied in 0.02 secs (39000 bytes/sec)
ftp>
```

f. After all files have been downloaded, exit the FTP session:

```
ftp> quit
```

g. At the C:/> prompt, type dir and verify all six client files are present on Laptop BR-1.

```
ftp> quit
221- Service closing control connection.
C:\> dir

Volume in drive C has no label.
Volume Serial Number is 5E12-4AF3
Directory of C:\

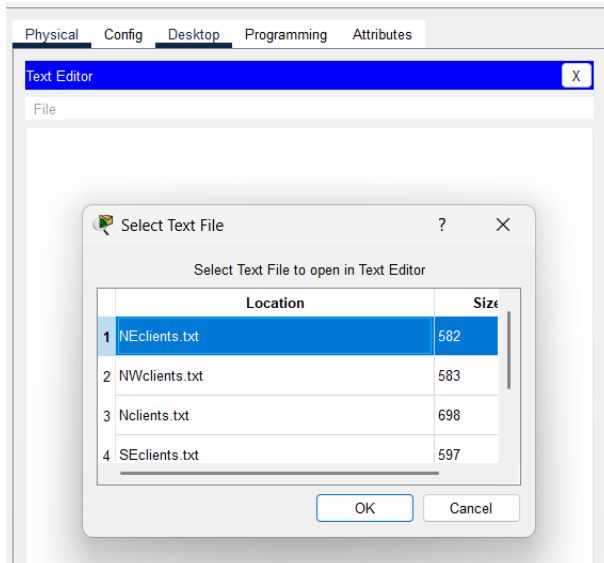
1/1/1970    2:0 PM           582      NEclients.txt
1/1/1970    2:0 PM           583      NWclients.txt
1/1/1970    2:0 PM           698      Nclients.txt
1/1/1970    2:0 PM           597      SEclients.txt
1/1/1970    2:0 PM           649      SWclients.txt
1/1/1970    2:0 PM           780      Sclients.txt
               3889 bytes          6 File(s)
C:\>|
```


Part 2: Use Hashing to Verify File Integrity

In this part, use the CSE-LABVM to compute MD5 hashes for each downloaded file. Compare each computed hash against the corresponding reference hash obtained in Part 1, Step 2. Any file whose hash does not match has been tampered with and must be escalated.

Step 1: Hash Each Client File and Compare Against Reference Values

- a. In Packet Tracer, close the Command Prompt on Laptop BR-1, then click Text Editor.
- b. Click File > Open, select NEclients.txt, then click OK.

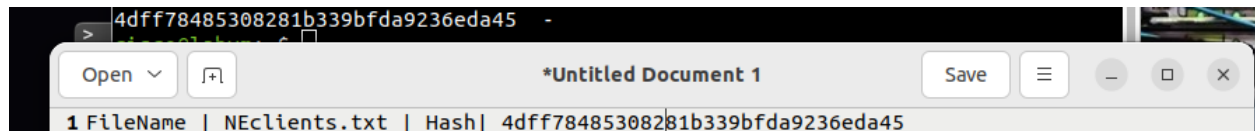


- c. Select all the document text (Ctrl+A) and copy it (Ctrl+C).
- d. Switch to the CSE-LABVM.
- e. Double-click the Terminal icon to open a terminal window.
- f. Run the following command to compute the MD5 hash. Replace 'file-contents-here' with the clipboard content (paste between the single quotes):

```
cisco@labvm:~$ echo -n 'file-contents-here' | md5sum
```

```
cisco@labvm: ~
File Edit View Search Terminal Help
cisco@labvm:~$ echo -n 'Name|Zip Code|Email|Pin #
Hannah Caldwell|7624|nec@ligula.net|6111
Rhannon B. Langley|46872|fermentum@curusnon.co.uk|9781
Nigel Ward|3584|convallis.erat.eget@luctusvulputatenisi.org|4896
Alvin Farley|69508|lectus@volutpat.co.uk|5358
Clark U. Pratt|23441|tincidunt.neque@nisl.co.uk|9273
Robin Randall|10108|faucibus.lectus.a@nonarcu.net|4232
Stacey L. Kirby|Y4B 8Z5|euismod.enim@mauris.org|4200
Joan Pearson|1867VB|convallis.erat@accumsannequeet.net|5002
Herman Lambert|09774|vitae.velit.egestas@tinciduntaliquamarcu.com|1220
Quentin Blankenship|48315-746|augue@consequat.edu|3387' | md5sum
4dff78485308281b339bfda9236eda45 -
cisco@labvm:~$
```

g. Compare the resulting hash value against the reference hash for NEclients.txt in your Pluma reference document.



h. Repeat steps b-g for each of the remaining five client files. Record all results in the table below.

Hash Comparison Table

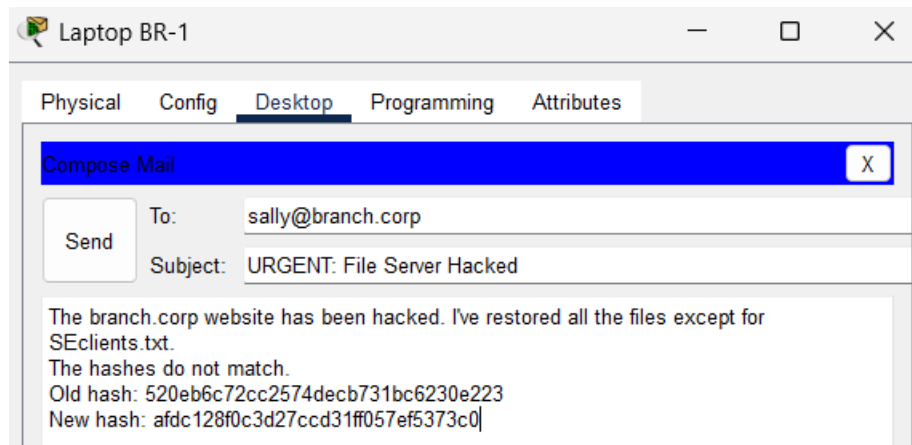
Filename	Reference Hash (MD5)	Computed Hash (MD5)	Match ?
NEclients.txt	4dff78485308281b339bfda9236eda45	4dff78485308281b339bfda9236eda45	Yes
NWclients.txt	2c77ff70d5f7a2a720d231617157faa4	a85eb70814f1d47c1274bb7d483c83fc	No
Nclients.txt	8040e56207aab9dd331566b79dca37dd	b65a3767afe4db203a52c8a50f87f434	No
SEclients.txt	520eb6c72cc2574dec731bc6230e223	afdc128f0c3d27ccd31ff057ef5373c0	No
SWclients.txt	69e751fe1175b39709629710e5ffa491	69e751fe1175b39709629710e5ffa491	Yes
Sclients.txt	06bea9b0fa7db92d7c21261b88323be6	06bea9b0fa7db92d7c21261b88323be6	Yes

Step 2: Escalate the Incident - Email Sally

a. Return to Packet Tracer and close the Text Editor on Laptop BR-1.

b. Click the Email application, then click Compose.

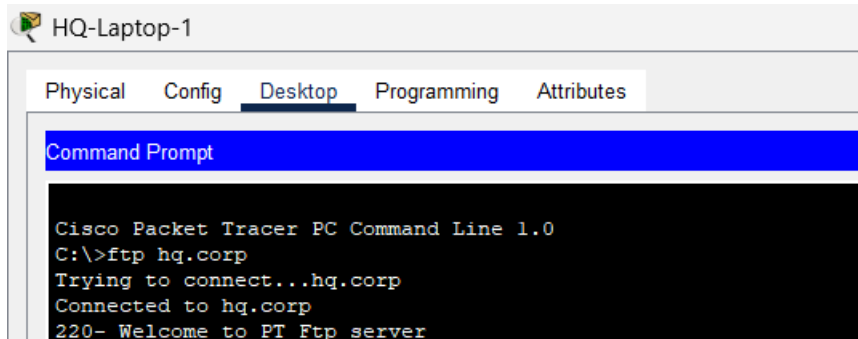
c. Address the email to sally@branch.corp. In the subject line write something like 'URGENT: File Server Hacked'. In the body, describe which file was found to have a mismatched hash and what action you believe should be taken. Click Send when complete.



Step 3: Download the Suspected File to Sally's PC

- In Packet Tracer, navigate to the HQ site and click HQ-Laptop-1 (Sally's PC).
- Click the Desktop tab > Command Prompt. Connect to the HQ FTP Server:

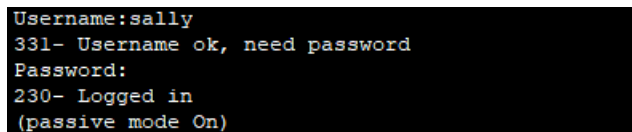
```
ftp hq.corp
```



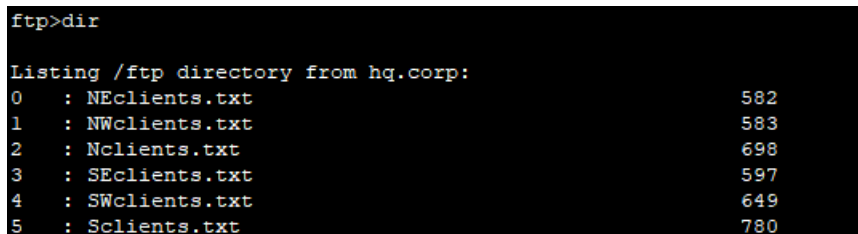
- Log in as Sally:

```
Username: sally
```

```
Password: cisco321
```

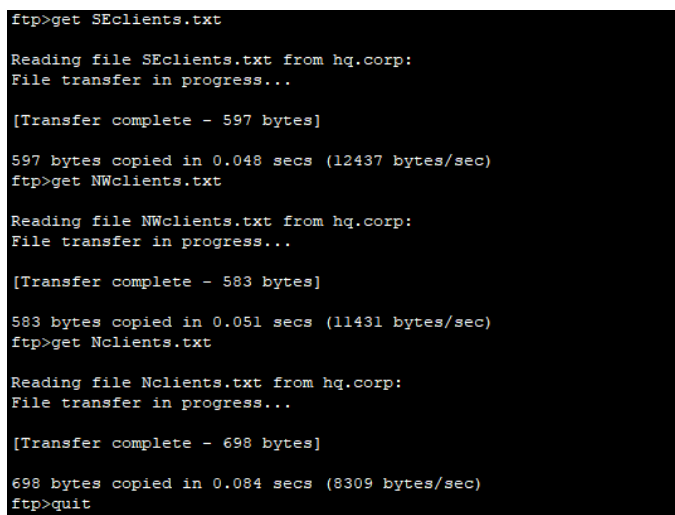


- At the ftp> prompt, enter dir to view the files on the remote HQ FTP Server.



- Download the tampered file identified in Step 1:

```
ftp> get <tampered-filename.txt>
```



f. Exit the FTP session:

```
ftp> quit
```

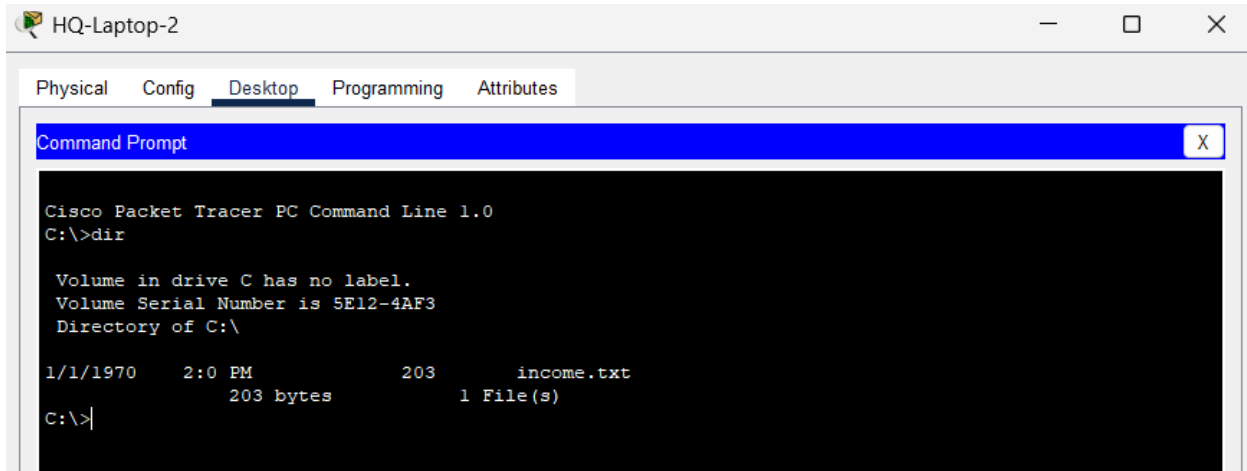
g. Enter dir at the C:/> prompt and verify the tampered file is now on HQ-Laptop-1.

```
221- Service closing control connection.  
C:\>dir  
  
Volume in drive C has no label.  
Volume Serial Number is 5E12-4AF3  
Directory of C:\  
  
1/1/1970    2:0 PM           583      NWclients.txt  
1/1/1970    2:0 PM           698      Nclients.txt  
1/1/1970    2:0 PM           597      SEclients.txt  
                1878 bytes      3 File(s)  
C:\>|
```

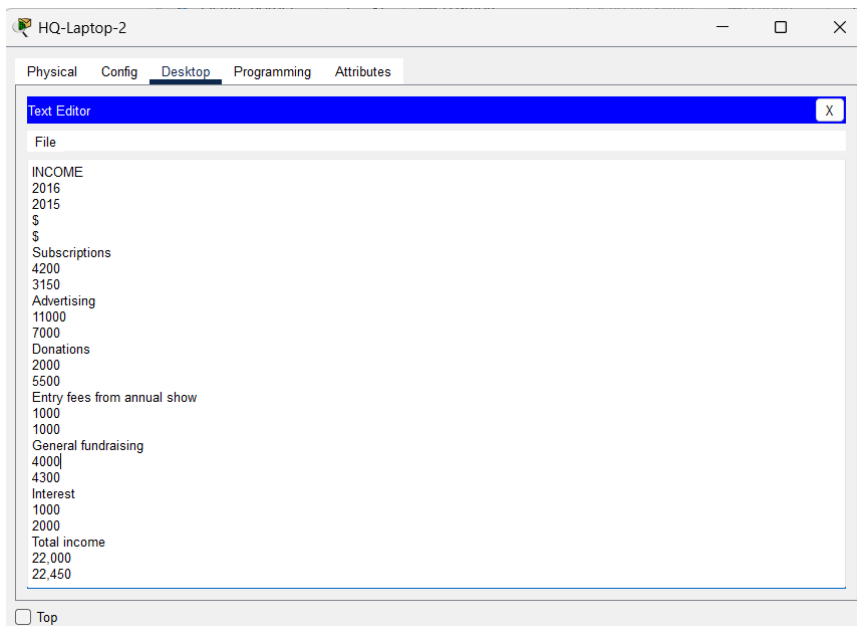
Part 3: Use HMAC to Verify File Integrity

Bob is the CFO for a small business and keeps track of all the company finances. In this part, you will compute and verify a Hash-based Message Authentication Code (HMAC) for a critical financial file to ensure it has not been altered. Unlike plain hashing, HMAC uses a secret key, providing both data integrity verification and sender authentication.

- Click Bob's laptop, which is HQ-Laptop-2.
- Click the Desktop tab > Command Prompt. Enter the `dir` command and verify that the file `income.txt` is present on the laptop. Close the command prompt when done.



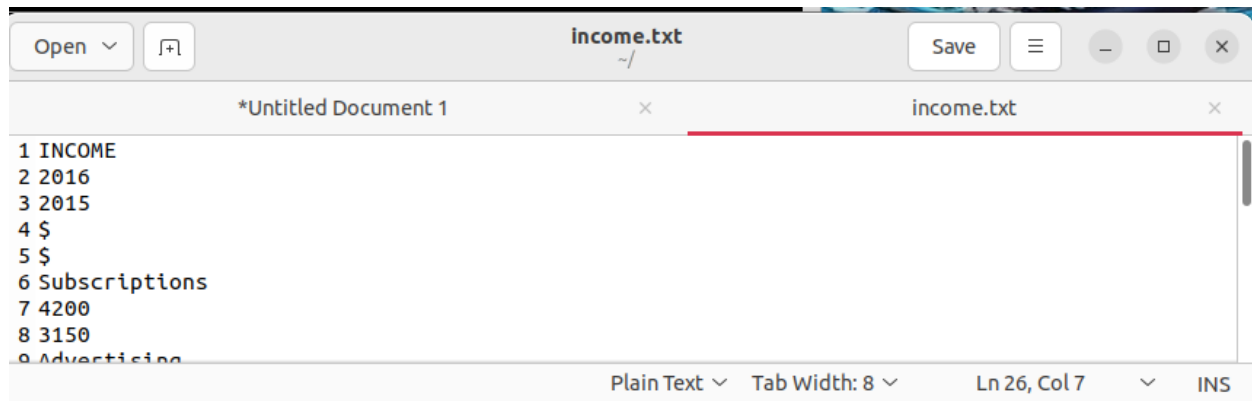
- Click Text Editor, then File > Open.
- Select the document `income.txt` and click OK.



- Select all document contents (`Ctrl+A`) and copy them (`Ctrl+C`).

f. Switch to the CSE-LABVM. Click Menu > Text Editor Pluma. Click Edit > Paste to paste the file contents into Pluma.

g. Click File > Save. Save the file with the name income.txt. Close Pluma when done.



h. Open a terminal window in the CSE-LABVM. Run the following OpenSSL command to generate an HMAC-SHA256 for income.txt using the secret key cisco123:

```
cisco@labvm:~$ openssl dgst -sha256 -hmac cisco123 income.txt
cisco@labvm:~$ openssl dgst -sha256 -hmac cisco123 income.txt
HMAC-SHA2-256(income.txt)= b138706cb55787d2a01934b224edad32203f87470ff6c7ffb9bd1
26786d1830d
```

Concepts & Analysis

MD5 (Message Digest 5)

Produces a 128-bit (32 hexadecimal character) hash. MD5 is fast and widely used for file integrity verification in non-adversarial contexts. However, it is cryptographically broken for collision resistance and should not be used for digital signatures, certificates, or password storage.

SHA-256 (Secure Hash Algorithm, 256-bit)

Part of the SHA-2 family, SHA-256 produces a 256-bit digest. It is significantly more collision-resistant than MD5 and is the current standard for TLS certificates, code signing, blockchain, and HMAC operations.

HMAC (Hash-based Message Authentication Code)

HMAC (RFC 2104) combines a cryptographic hash function with a secret key. It provides both data integrity and sender authentication. An attacker who intercepts a message cannot forge a valid HMAC without knowing the secret key, making it far more secure than plain hashing for authentication purposes.

Avalanche Effect

A property of cryptographic hash functions: a small change in the input (even a single bit) produces a drastically different output hash. This makes hashes effective for detecting file tampering because any modification, no matter how minor, results in a completely different hash value.

File Integrity Monitoring (FIM)

A security technique that uses hashes to detect unauthorized changes to critical files. FIM is required by compliance frameworks such as PCI-DSS (Requirement 11.5) and HIPAA. Enterprise tools include Tripwire, OSSEC, and Wazuh.

FTP Security Considerations

FTP transmits credentials and data in plaintext, making it vulnerable to eavesdropping and man-in-the-middle attacks. In production environments, SFTP or FTPS should be used. The credentials used in this lab (cisco123, cisco321) are demonstration-only and represent weak password practices that should never be used in real environments.

Conclusion

- Hashing provides a reliable mechanism for detecting unauthorized file modifications. The avalanche effect ensures that even the smallest input change produces a completely different hash value.
- By archiving reference hashes prior to an attack, administrators can quickly identify tampered files by re-computing and comparing hashes after recovery.
- HMAC extends plain hashing by incorporating a secret key, providing both integrity verification and sender authentication - making it more secure for protecting critical business files.
- The lab demonstrated a realistic incident-response workflow: Detect (hash mismatch) -> Investigate (identify tampered file) -> Escalate (email Sally) -> Contain (transfer file to analyst's PC for analysis).
- FTP is unsuitable for production use as it transmits data in cleartext. Real-world deployments must use encrypted transfer protocols such as SFTP or FTPS.

References

1. Cisco Networking Academy. (2022). Packet Tracer - File and Data Integrity Checks. Cisco Systems.
2. NIST. (2012). SP 800-107 Rev. 1: Recommendation for Applications Using Approved Hash Algorithms. National Institute of Standards and Technology.
3. Krawczyk, H., Bellare, M., & Canetti, R. (1997). HMAC: Keyed-Hashing for Message Authentication. RFC 2104. Internet Engineering Task Force (IETF).
4. OpenSSL Foundation. (2024). openssl-dgst - perform digest operations.
5. OWASP. (2023). Cryptographic Storage Cheat Sheet. OWASP Foundation.